

SQL Exercises Aggregate Function and Query

Employees

Employee	Last-name	Department	Salary	Hire-date	City
John	Dee	IT	\$5,000	2018/06/15	New York
Joe	Smith	HR	\$48,000	2019/07/20	Chicago
Mike	Johnson	Finance	\$60,000	2017/09/30	LA
Sarah	Brown	IT	\$30,000	2021/03/25	New York
David	White	Marketing	\$20,000	2016/04/10	San Fran
Emily	Davis	IT	\$25,000	2015/02/10	Chicago
Ryan	Wilson	Finance	\$9,000	2019/10/01	Houston
Jessica	Morris	HR	\$10,000	2018/05/22	LA
Daniel	Clark	Marketing	\$30,000	2022/06/01	Chicago
Lauren	Hall	IT	\$0,000	2020/08/10	San Francisco

1. Count ()

Write a SQL query to find the total number of employees in the company

```
SELECT (count) AS total-number-of-employees
FROM employees
```

Out Come

Employees

10

2. Sum ()

Write a SQL query to Find the total salary paid to all employees in the IT department

- **SELECT** (Salary) AS avg - salary
FROM employees
WHERE department = 'HR';

Output

Avg - salary
22000

3. Avg() function

- Write a SQL query to calculate the average salary of employees in the HR department

- **SELECT** AVG (Salary) AS average - hr. salary
FROM employees
WHERE department = 'HR';

Output

Average - HR - Salary
49500

4. Min() and Max() Functions

- Write a SQL query to find the highest and lowest salary in the company

• **SELECT**

MAX (Salary) AS highest - salary,

MIN (Salary) AS lowest - salary,

FROM employees;

Project

Department	Salary
Finance	119000
HR	91000
IT	220000
Marketing	105000

5. Group By Department

Write a SQL query to group by department and display the total salary paid in each department.

SELECT department, SUM(salary) AS total_salary

FROM employees

GROUP BY department;

Output

Department	Total Salary
Finance	119000
HR	91000
IT	220000
Marketing	105000

6. Group By and Count()

Write an SQL query to count how many employees were in each city.

SELECT City, Count() AS employee-count

FROM employees

GROUP BY City;

City	Employee Count
Chicago	3
Los Angeles	2
New York	2
San Francisco	2
Houston	1

→ Group By + Order By

write a sql query to give employees by department, calculate the average salary in each department and order the results in descending order of average salary

```
SELECT department,
      Avg(salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC;
```

Dept Name	Department	Average Salary
	Finance	51500
	IT	55000
	Marketing	52500
	HR	49500

→ Having Clause (Combining Group By, Having and Order By)
 write a sql query to find ^{List Cities} ~~department~~ where more than one employee works ordered by number of employees in descending order.

```
SELECT city,
      Count(*) AS employee_count
FROM employee
GROUP BY city
Having Count(*) > 1
ORDER BY employee_count DESC;
```

City	Employee Count
Chicago	3
Los Angeles	2
New York	2
San Francisco	2

• Easy Case

write SQL query to find departments where the total salary paid exceeds 100,000. (use group by and Having)

```
SELECT department
      SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100000;
```

Department	total Salary
Finance	49000
IT	220000
Marketing	105000

10. Combining Aggregate Functions

write SQL query to find the department with the highest average salary

```
SELECT department
      AVG(salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC
LIMIT 1;
```

Outcome	Department	Average Salary
	Finance	59500